SMART IRRIGATION ADVISORY

Statistical Analysis & Engineering Decision Report

DATA-DRIVEN • WATER-AWARE • ACTIONABLE

2,200	3	3
DATASET RECORDS	**MEASURED INPUTS**	**ADVISORY OUTPUTS**
all rows retained	temperature, rainfall, moisture	irrigate, monitor, no irrigation

01 / INTRODUCTION

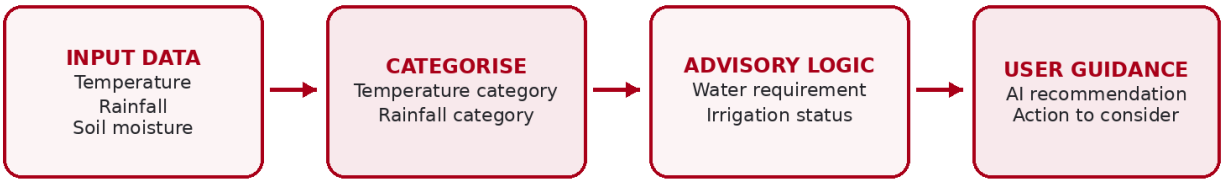
The project converts temperature, rainfall and soil-moisture measurements into simple irrigation guidance. The final dataset includes five derived fields: temperature category, rainfall category, water requirement, irrigation status and AI recommendation.

02 / PROBLEM STATEMENT

Fixed schedules or guesswork can lead to overwatering or insufficient watering. The project aims to support timely decisions by combining environmental measurements and presenting a clear advisory instead of raw numbers alone.

HOW THE DATA BECOMES GUIDANCE

DATA-TO-ADVISORY WORKFLOW



Original 23 columns → 3 retained measurements + 5 derived fields → 8-column final dataset

Figure 1. Dataset processing flow. Derived labels make the measurements easier to interpret; they do not, by themselves, prove recommendation accuracy.

DATA PREPARATION The dataset was reduced from 23 to 8 columns: 3 measured fields were retained and 5 advisory fields were added.

03 / INTERPRETATION OF RESULTS

The dataset has 2,200 rows, eight columns and no missing values in the supplied CSV. Units and the origin of the records are not specified.

25.62	103.46	20.15
MEAN TEMPERATURE units not specified	**MEAN RAINFALL** units not specified	**MEAN SOIL MOISTURE** SD = 5.79

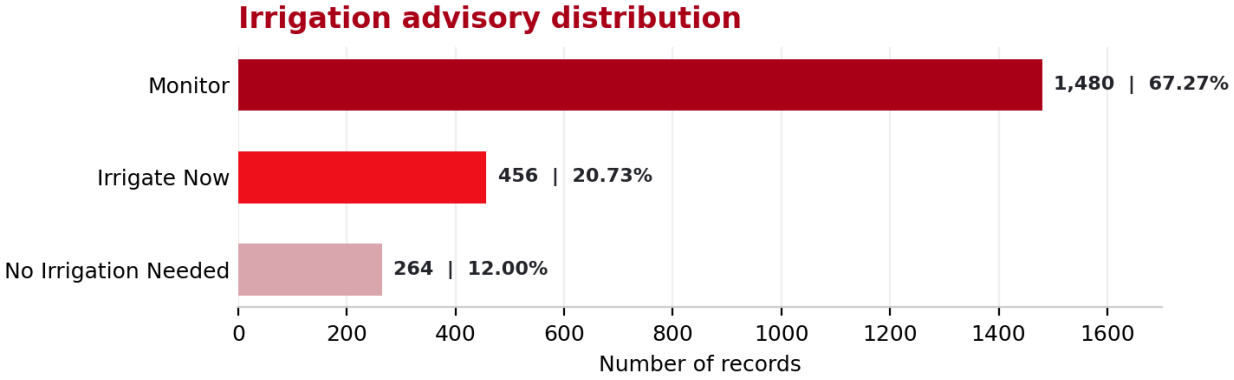
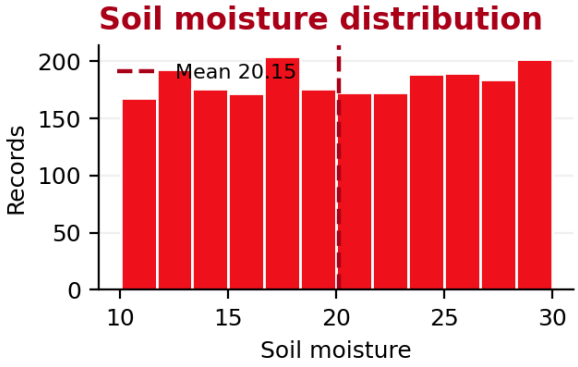
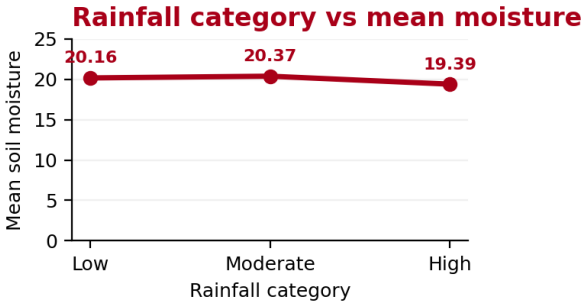


Figure 2. Advisory labels across all 2,200 records.





Figures 3–4. Soil-moisture distribution and mean moisture by rainfall category.

WHAT THE GRAPHS SHOW

Monitor is the largest group: 1,480 records (67.27%). “Irrigate Now” accounts for 456 records (20.73%).

WHAT THEY DO NOT PROVE

Mean soil moisture is 20.16 (low rainfall), 20.37 (moderate) and 19.39 (high). Correlations with moisture are very weak: rainfall r = -0.049; temperature r = -0.010.

Interpretation: the status distribution describes the dataset’s generated labels. Because these labels are derived within the same dataset, the patterns are not an independent test of prediction accuracy.

04 / AI & ENGINEERING DECISION

The proposed advisory logic should prioritise soil moisture, then check recent or forecast rainfall. Temperature adds context because hotter conditions can increase water loss.

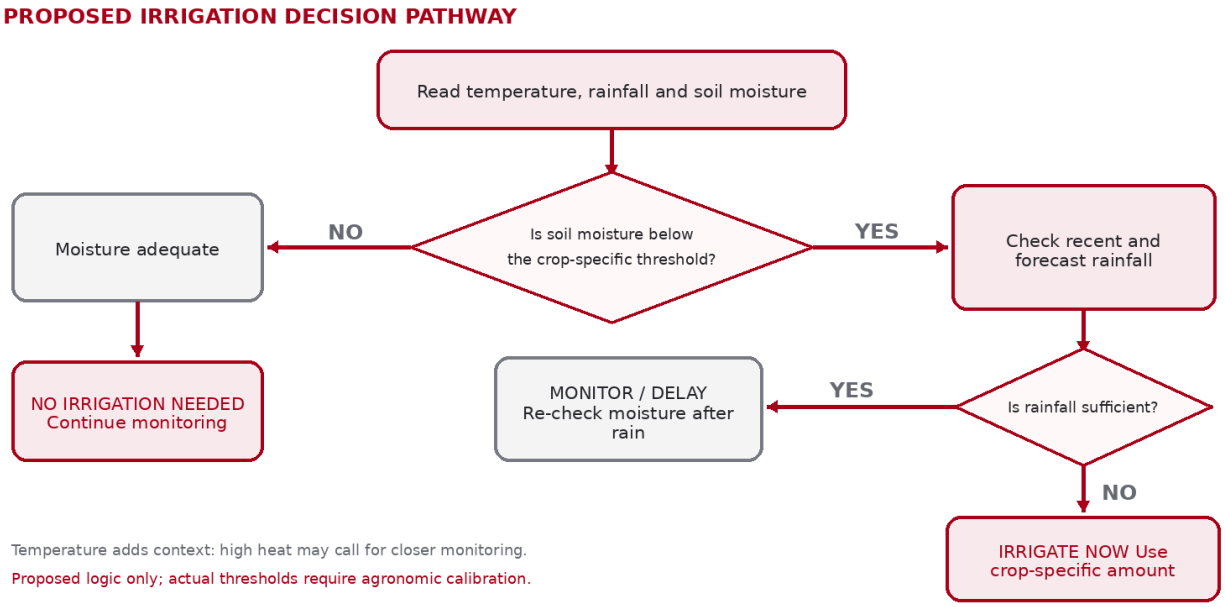


Figure 5. Recommended decision pathway for a future implementation. It is not a verified diagram of the live website's internal code.

A useful next step is to calibrate thresholds for the crop and soil, then validate recommendations against field measurements.



Figure 6. Validation loop: real-world outcomes should feed back into testing and improvement.

05 / LIMITATIONS

Reduced feature set Soil type, humidity, crop label, growth stage and other original fields were removed.	**Assumed thresholds** Exact rules for categories, water requirement and irrigation status are not documented.
Unknown provenance The CSV does not identify whether records are sensor, historical or simulated data; units are unstated.	**No field validation** Water savings, crop outcomes and recommendation accuracy have not been established.

06 / CONCLUSION

The project demonstrates how environmental data can be converted into understandable irrigation advice. Results are promising as a decision-support prototype, but reliable thresholds, crop-specific inputs and field validation are needed before claiming practical accuracy or water savings.